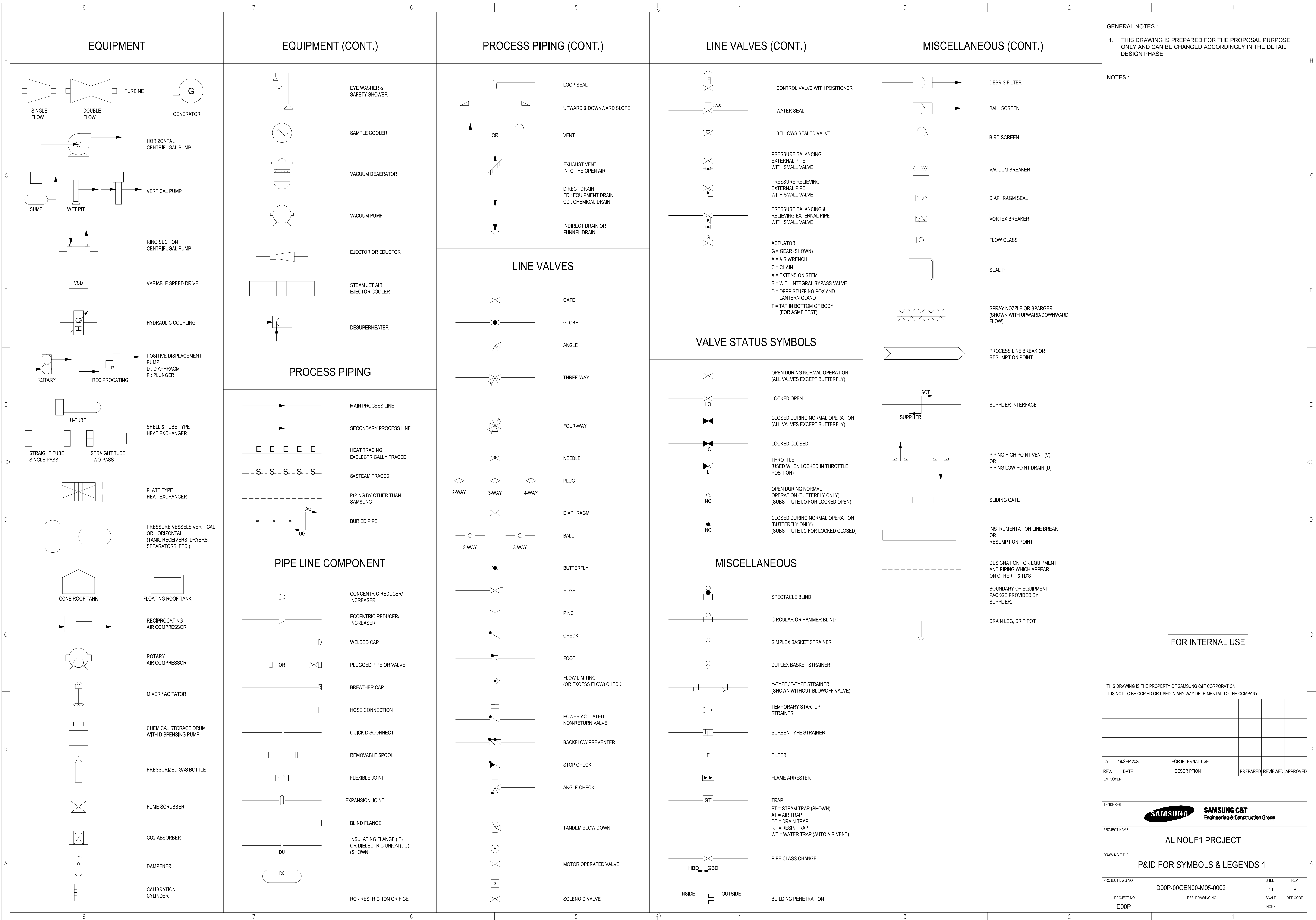
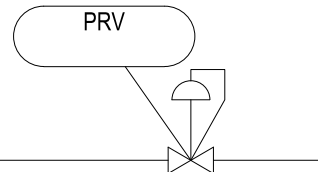
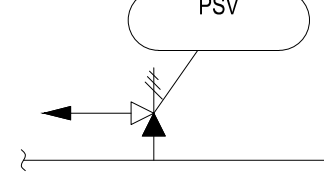
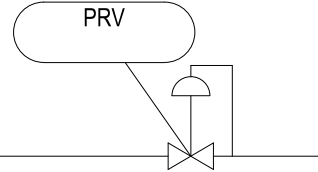
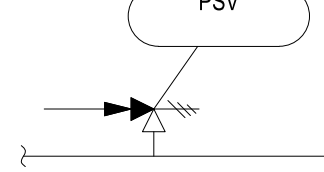
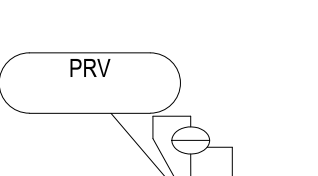
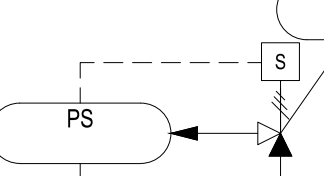
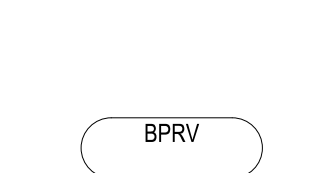
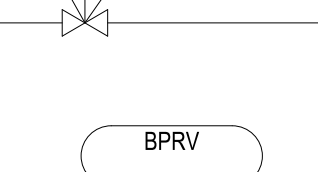
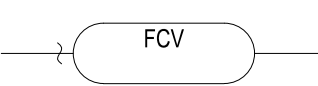
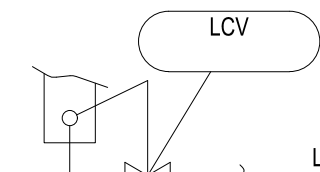
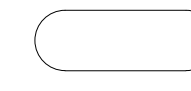
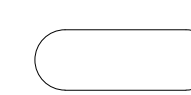
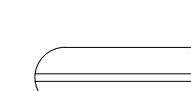
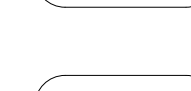
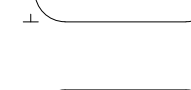
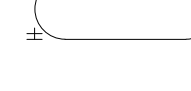
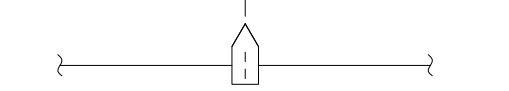
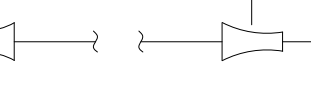
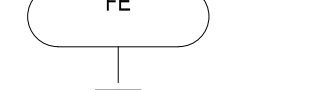
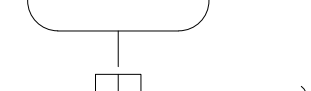
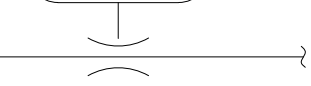
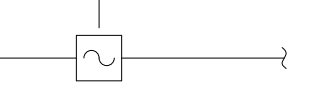
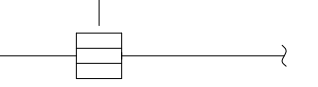
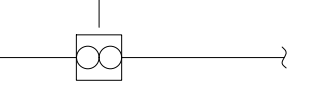
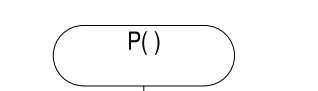
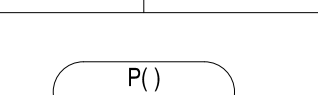
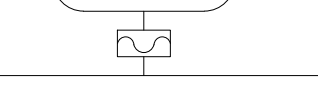
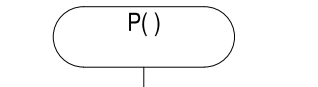
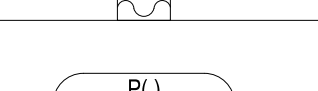
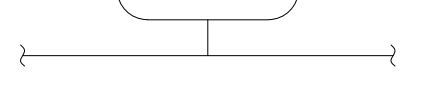
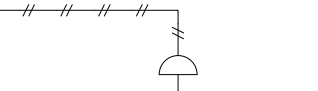
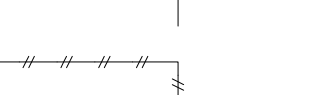
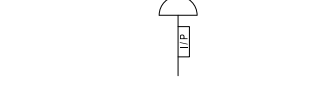
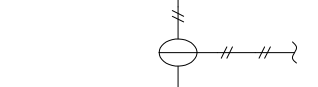
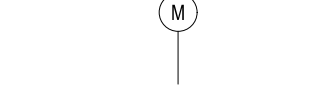
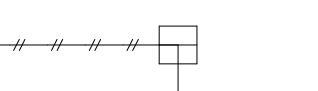
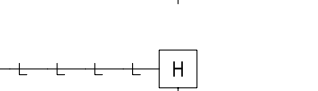
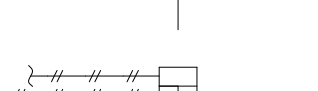
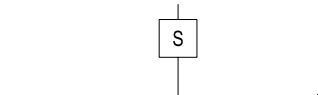
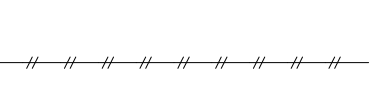
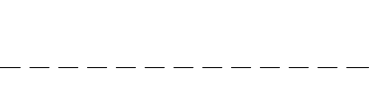
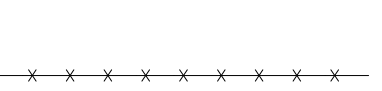
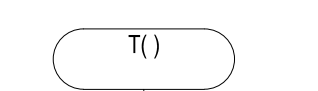
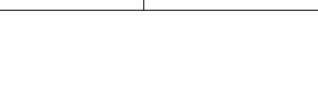
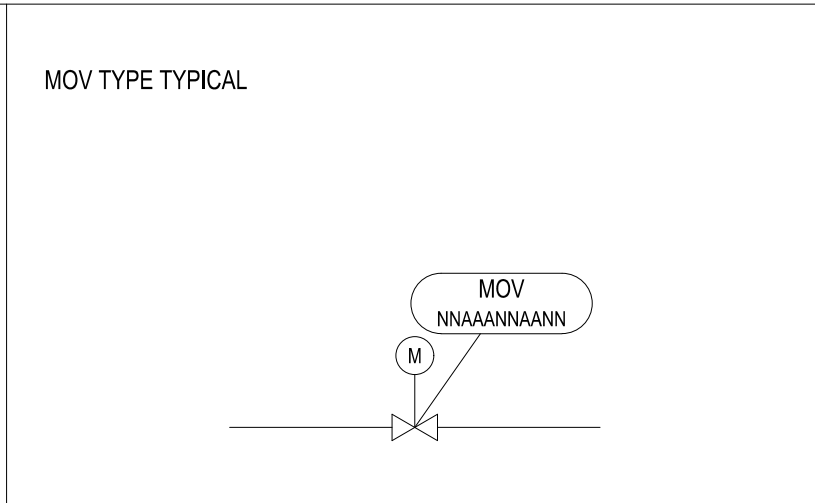
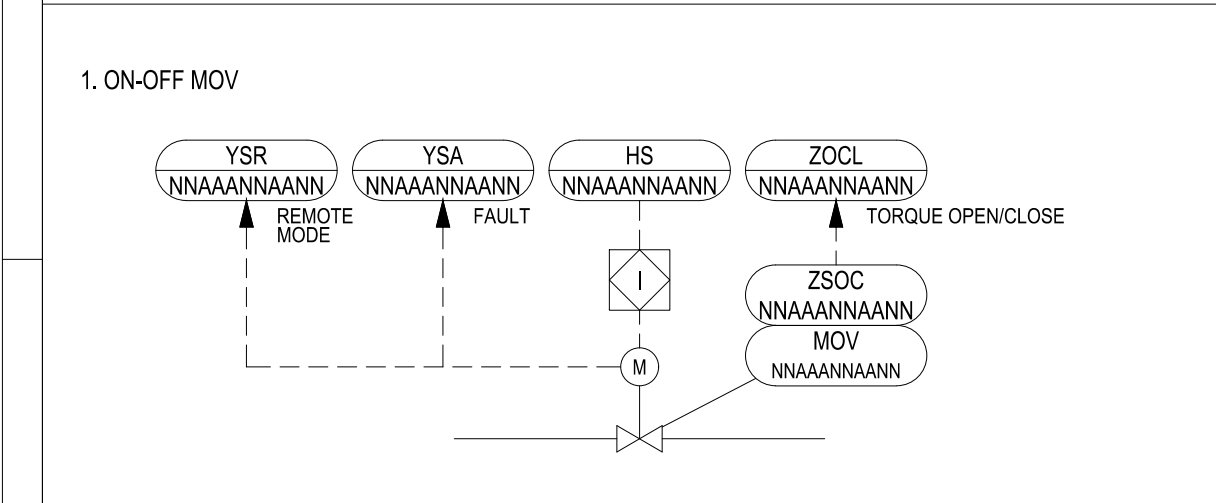




8		7		6		5		4		3		2		1									
FIRST LETTER		SUCCEEDING LETTERS																		SELF-ACTUATED DEVICES-PRESSURE		GENERAL NOTES : 1. THIS DRAWING IS PREPARED FOR THE PROPOSAL PURPOSE ONLY AND CAN BE CHANGED ACCORDINGLY IN THE DETAIL DESIGN PHASE.	
		SENSING DEVICE		DISPLAY DEVICE				CONTROL DEVICE				MISCELLANEOUS DEVICE											
SYMBOL	MEASURED OR INITIATING VARIABLE	PRIMARY ELEMENT	TRANSMITTER	INDICATOR	RECORDER	INTEGRATING INDICATOR	ARARM ANNUNCIATOR			CONTROL STATION MAN/AUTO	CONTROLLER	CONTROL VALVE	OTHER FINAL CONTROL ELEMENT	SWITCH	LOCAL OBSERVATION GLASS	SOLENOID RELAY OR POSITIONER	TEST POINT	USER'S CHOICE	UNCLASSIFIED	PRESSURE REDUCING REGULATOR SELF-CONTAINED PRESSURE RELIEF OR SAFETY VALVE ANGLE PATTERN SPRING OR WEIGHT LOADED OR WITH INTEGRAL PILOT VALVE OR WITH INTEGRAL PILOT VALVE PRESSURE REDUCING REGULATOR WITH EXTERNAL PRESSURE TAP VACUUM RELIEF VALVE ANGLE PATTERN SPRING OR WEIGHT LOADED OR WITH INTEGRAL PILOT VALVE DIFFERENTIAL PRESSURE REDUCING REGULATOR WITH INTERNAL AND EXTERNAL PRESSURE TAPS WITH INTERNAL AND EXTERNAL PRESSURE TAPS PRESSURE RELIEF OR SAFETY VALVE BY INTEGRAL SOLENOID BACKPRESSURE REGULATOR SELF-CONTAINED RUPTURE DISC OR SAFETY HEAD FOR PRESSURE RELIEF BACKPRESSURE REGULATOR WITH EXTERNAL PRESSURE TAP RUPTURE DISC OR SAFETY HEAD FOR VACUUM RELIEF SELF-ACTUATED DEVICES-FLOW SELF-ACTUATED DEVICES-LEVEL FLOW REGULATOR LEVEL REGULATOR WITH MECHANICAL LINKAGE			
()	TYPICAL SYMBOL	()E	()T	()I	()R	()QI	()AL	()AH	()AHL	()K	()C	()CV	()Z	()S	()G	()Y	()P	()N	()X				
A	ANALYSIS	A E	A T	A I	A R		AAL	AAH	AAHL	A K	A C		A Z	A S		A Y	A P	A N	A X				
B	BURNER FLAME	B E	B T	B I	B R		BAL				B C			B S		B Y	B P	B N	B X				
C	CONDUCTIVITY	C E	C T	C I	C R		CAL	CAH	CAHL	C K			C Z	C S		C Y	C P	C N	C X				
D	DENSITY (MASS) OR SPECIFIC GRAVITY	D E	D T	D I	D R		DAL	DAH	DAHL	D K	D C		D Z	D S		D Y	D P	D N	D X				
E	VOLTAGE (EMF)	E E	E T	E I	E R		EAL	EAH	EAHL	E K	E C		E Z	E S		E Y		E N	E X				
F	FLOW	F E	F T	F I	F R	FQI	FAL	FAH	FAHL	F K	F C	FCV	F Z	F S	F G	F Y	F P	F N	F X				
G	GAUGING (DIMENSIONAL)	G E	G T	G I	G R		GAL	GAH	GAHL	G K	G C			G S		G Y		G N					
H	HAND (MANUAL INITIATED)	H									H C			H S		H Y		H N					
I	CURRENT	I E	I T	I I	I R	IQI	IAL	IAH	IAHL	I K			I Z	I S		I Y		I N	I X				
J	POWER	J E	J T	J I	J R	JQI	JAL	JAH	JAHL	J K			J Z	J S		J Y		J N	J X				
K	TIME	K E	K T	K I	K R	KQI	KAL	KAH	KAHL		K C		K Z	K S		K Y		K N	K X				
L	LEVEL	L E	L T	L I	L R		LAL	LAH	LAHL	L K	L C	LCV	L Z	L S	L G	L Y	L P	L N	L X				
M	MOISTURE OR HUMIDITY	M E	M T	M I	M R		MAL	MAH	MAHL	M K	M C		M Z	M S		M Y	M P	M N	M X				
O	USER'S CHOICE																						
P	PRESSURE OR VACUUM	P E	P T	P I	P R		PAL	PAH	PAHL	P K	P C	PCV	P Z	P S		P Y	P P	P N	P X				
PD	PRESSURE DIFFERENTIAL	PDE	PDT	PDI	PDR		PDAL	PDAH	PDAHL	PDK	PDC		PDZ	PDS									
Q	QUANTITY	Q E	Q T	Q I	Q R	QQI	QAL	QAH	Q AHL	Q K	Q C		Q Z	Q S		Q Y		Q N	Q X				
S	SPEED OR FREQUENCY	S E	S T	S I	S R	SQI	SAL	SAH	SAHL	S K	S C		S Z	S S		S Y		S N	S X				
T	TEMPERATURE	T E	T T	T I	T R		TAL	TAH	TAHL	T K	T C	TCV	T Z	T S		T Y	T P	T N	T X				
U	MULTIVARIABLE			U I	U R								U Z	U S		U Y		U N	U X				
V	VIBRATION		V T	V I	V R		VAL	VAH	VAHL	V K	V C		V Z	V S		V Y	V P	V N	V X				
W	WEIGHT OR FORCE	W E	W T	W I	W R	WQI	WAL	WAH	WAHL	W K	W C		W Z	W S		W Y		W N	W X				
X	UNCLASSIFIED																						
Y	EVENT OR STATE	Y E	Y T	Y I	Y R		YAL	YAH					Y Z	Y S		Y Y		Y N	Y X				
Z	POSITION	Z E	Z T	Z I	Z R		ZAL	ZAH	Z AHL				Z Z	Z S		Z Y		Z N	Z X				
GENERAL INSTRUMENTS				SENSOR-FLOW				SENSOR-FLOW (CONT.)				SENSORS-PRESSURE				VALVES ACTUATORS							
FF INSTRUMENT FOR SINGLE MEASURED VARIABLE AND ANY NUMBER OF FUNCTIONS. INSTRUMENT IS LOCALLY MOUNTED.  INSTRUMENT FOR SINGLE MEASURED VARIABLE AND ANY NUMBER OF FUNCTIONS. INSTRUMENT IS LOCALLY MOUNTED. FF=FOUNDATION FIELDBUS  INSTRUMENT IS MOUNTED ON A LOCAL PANEL (TYPICAL)  INSTRUMENT IS MOUNTED ON A MAIN CONTROL PANEL (TYPICAL)  INSTRUMENT FOR PERFORMANCE TEST ONLY  INSTRUMENT FOR PERFORMANCE TEST PLUS NORMAL USE  INSTRUMENT IS MOUNTED BEHIND A PANEL (TYPICAL)				 FE - ORIFICE PLATE  ORIFICE PLATE IN QUICK - CHANGE FITTING  VENTURI TUBE / FLOW NOZZLE  PITOT TUBE OR PITOT VENTURI TUBE  FORWARD-REVERSE PITOT TUBE  FLUME  VORTEX  TURBINE OR PROPELLER TYPE  ROTAMETER  IN LINE INSTRUMENT				 ULTRASONIC  FLOW STRAIGHTENING VANES  POSITIVE DISPLACEMENT TYPE  DIS - DISPLACEMENT (SHOWN) BNN - BNNNETIC DP - DIFFERENTIAL PRESSURE FLO - FLOAT REF - REFLEX FLAT GLASS  DP - DIFFERENTIAL PRESSURE (SHOWN) BNN - BNNNETIC  GWR - GUIDED WAVE RADAR (SHOWN) RAD - RADAR FLO - FLOAT DIS - DISPLACEMENT				 DIRECT CONNECTED  WITH DIAPHRAGM SEAL PIPED  WITH DIAPHRAGM SEAL LINE OR EQUIP MOUNTED  STRAIN-GAGE TYPE  TW - THERMOWELL TE - TEMPERATURE ELEMENT WITH THERMOWELL TIT - TEMPERATURE TRANSMITTER WITH THERMOWELL AND TEMPERATURE ELEMENT  DUAL SENSOR (SHOWN WHEN BOTH ARE USED) RTD-RESISTANCE TEMPERATURE DETECTOR K- K type THERMO COUPLE  FILLED SYSTEM				 PNEUMATIC DIAPHRAGM  PNEUMATIC DIAPHRAGM WITH POSITIONER  DIAPHRAGM PRESSURE-BALANCED  ROTARY MOTOR (SHOWN TYPICALLY WITH ELECTRIC SIGNAL)  PNEUMATIC CYLINDER SINGLE-ACTING  HYDRAULIC CYLINDER SINGLE-ACTING  PNEUMATIC CYLINDER DOUBLE-ACTING  PNEUMATIC CYLINDER DOUBLE ACTING WITH PILOT VALVE  SINGLE SOLENOID (NOTE 14) RESET AFTER LATCH (OPTIONAL)  ELECTRO OR HYDRAULIC  HAND ACTUATOR (MOUNTED AT THE TOP SIDE OR BOTTOM OF ACTUATED DEVICE AS APPLICABLE)  UNCLASSIFIED (TYPE OF ACTUATOR TO BE WRITTEN ADJACENT TO THE SYMBOL)							
SIGNALS & LINES				SENSOR-LEVEL				SENSOR-TEMPERATURE				VALVE BODIES & FAILURE MODES											
 CONNECTION TO PROCESS OR MECHANICAL LINE OR INSTRUMENT SUPPLY  PNEUMATIC SIGNAL OR UNDEFINED SIGNAL FOR PROCESS FLOW DIAGRAMS  ELECTRIC SIGNAL  CAPILLARY TUBE (FILLED SYSTEM)  HYDRAULIC SIGNAL  ELECTROMAGNETIC OR SONIC SIGNAL  SOFTWARE OR DATA LINK				 THROUGH-FLOW TYPE 				 TWO - WAY (TYPICAL GLOBE SHOWN) FO - INDICATES FAIL OPEN FC - INDICATES FAIL CLOSED FL - INDICATES LOCKED (POSITION DOES NOT CHANGE) FI - INDICATES FAIL INDETERMINATE  THREE - WAY															
8		7		6		5		4		3		2		1									

	8	7	6
H	VALVE BODY WITH ACTUATOR-TYPICAL FOR MOTOR OPERATED VALVES		



2. INCHING

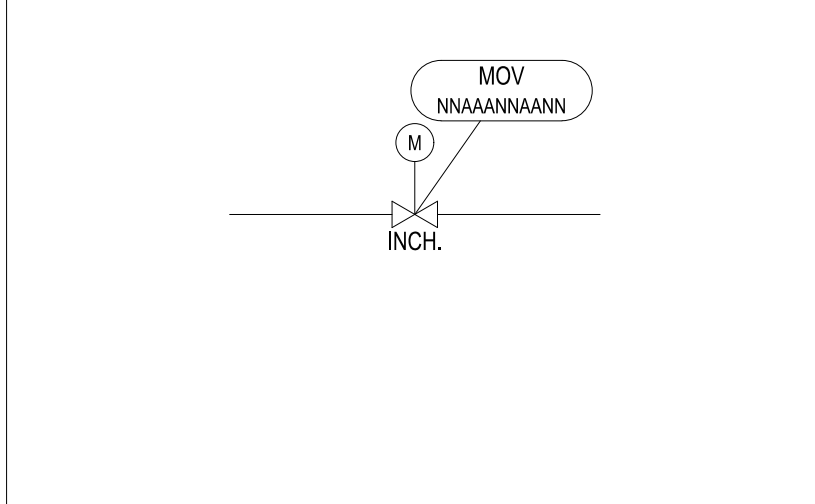
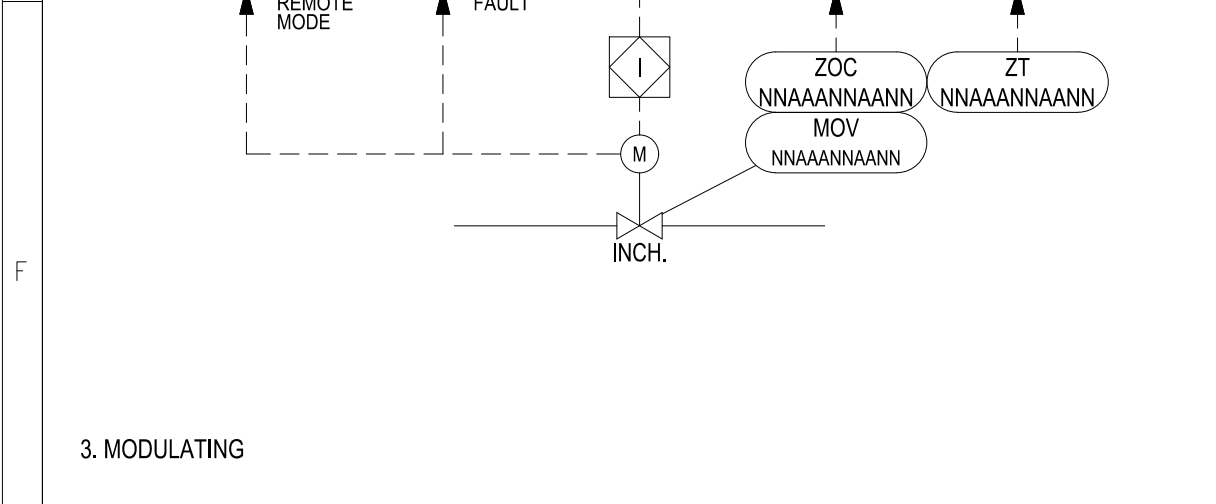
YSR
NNA¹AAANNA¹AA¹NN

YSA
NNA¹AAANNA¹AA¹NN

ZIC
NNA¹AAANNA¹AA¹NN

ZOCL
NNA¹AAANNA¹AA¹NN

ZI
NNA¹AAANNA¹AA¹NN



3. MODULATING

YSR
NNAAANNAANN

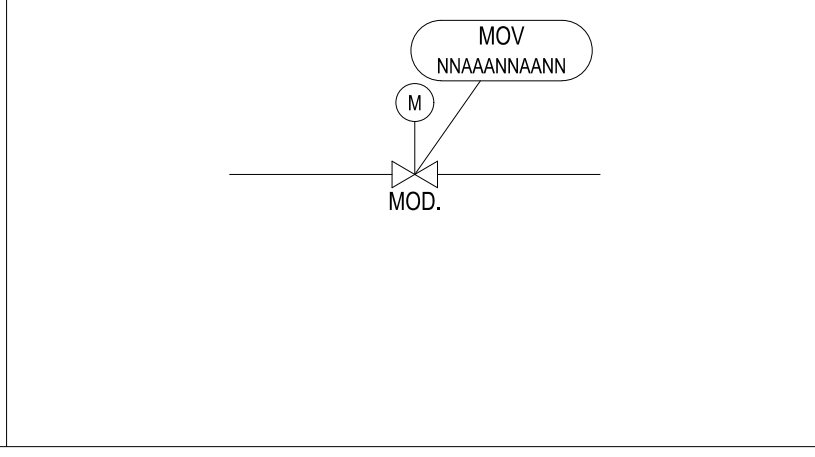
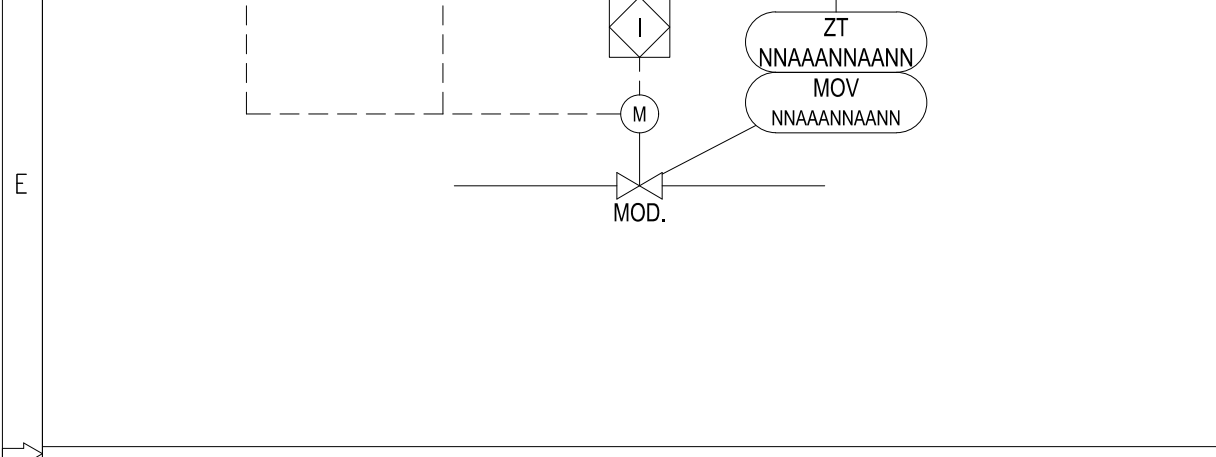
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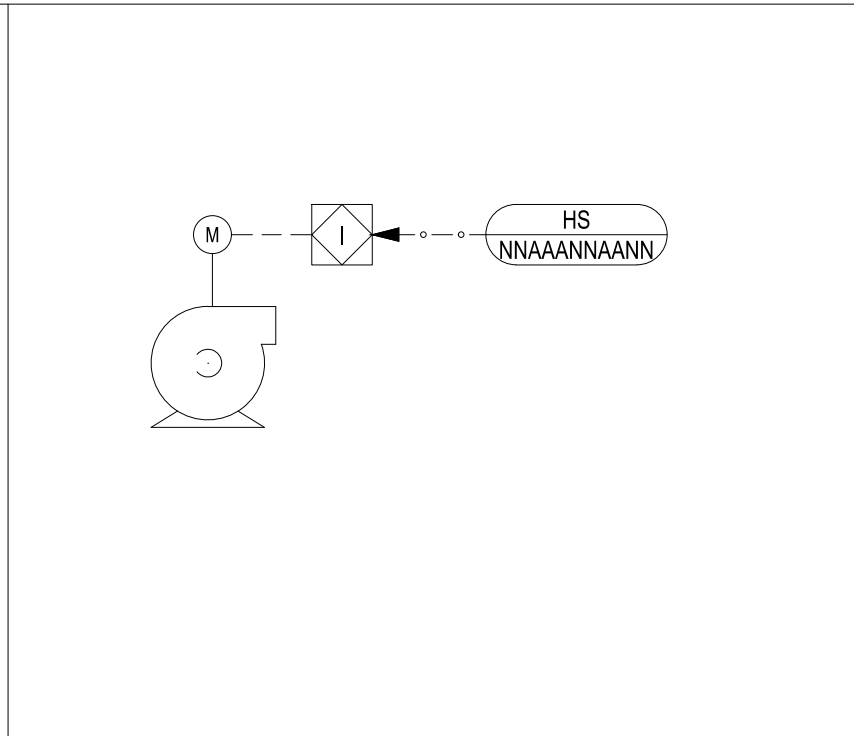
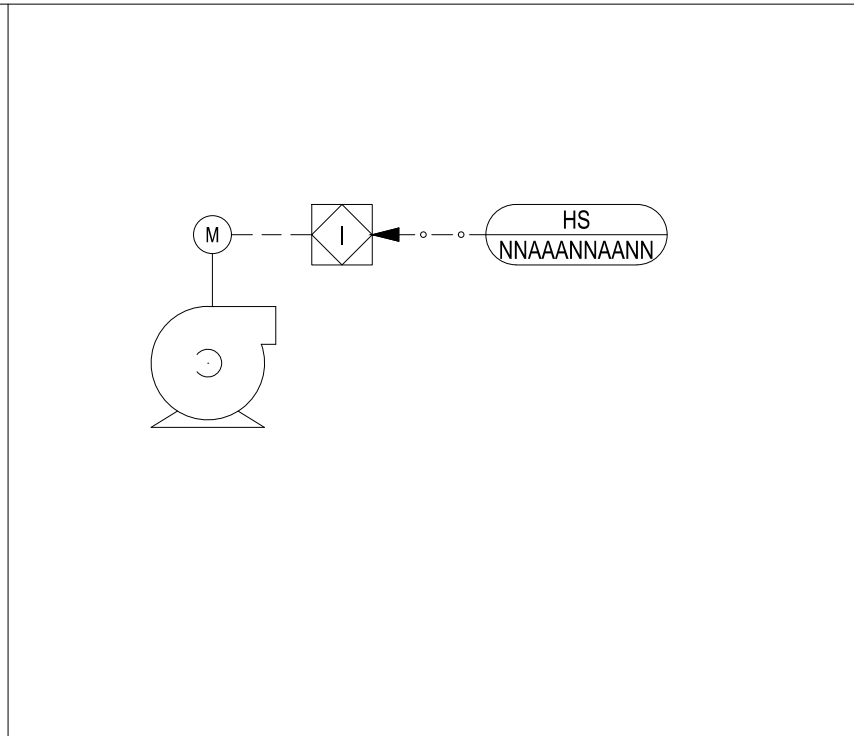
ZI
NNAAANNAANN

REMOTE MODE

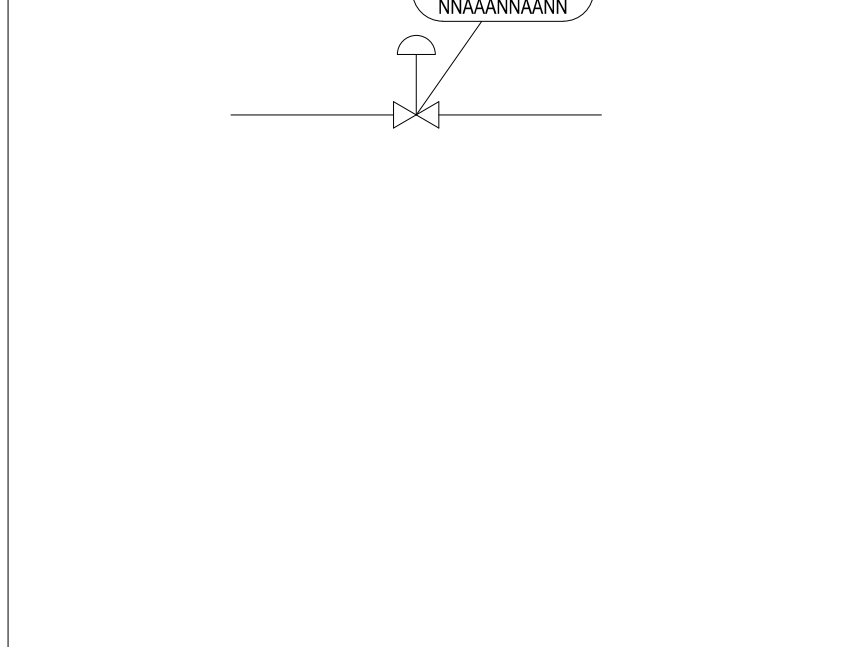
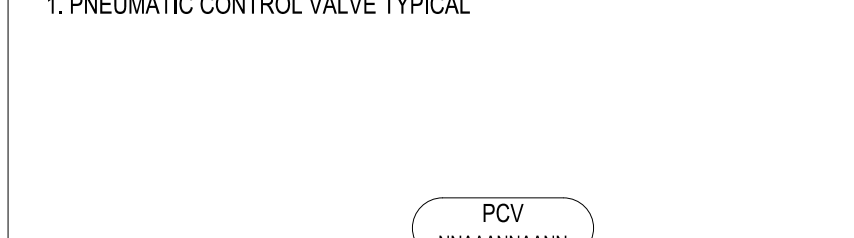
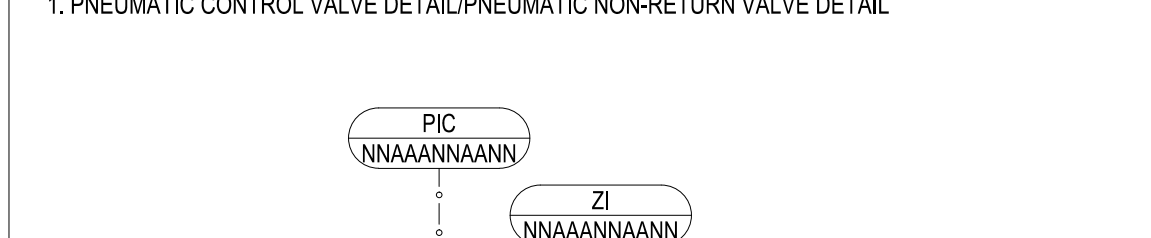
FAULT



MOTOR WITHOUT DUTY/STANDBY SELECTION (TYP.1)



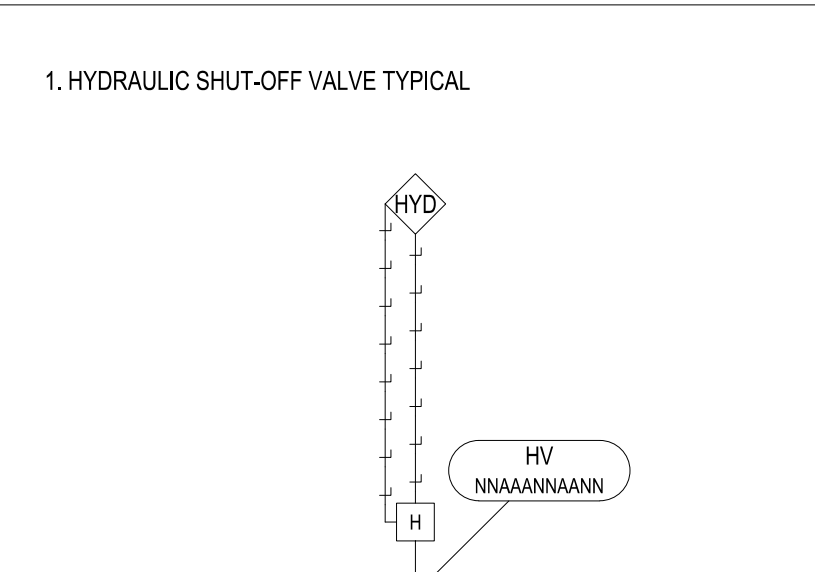
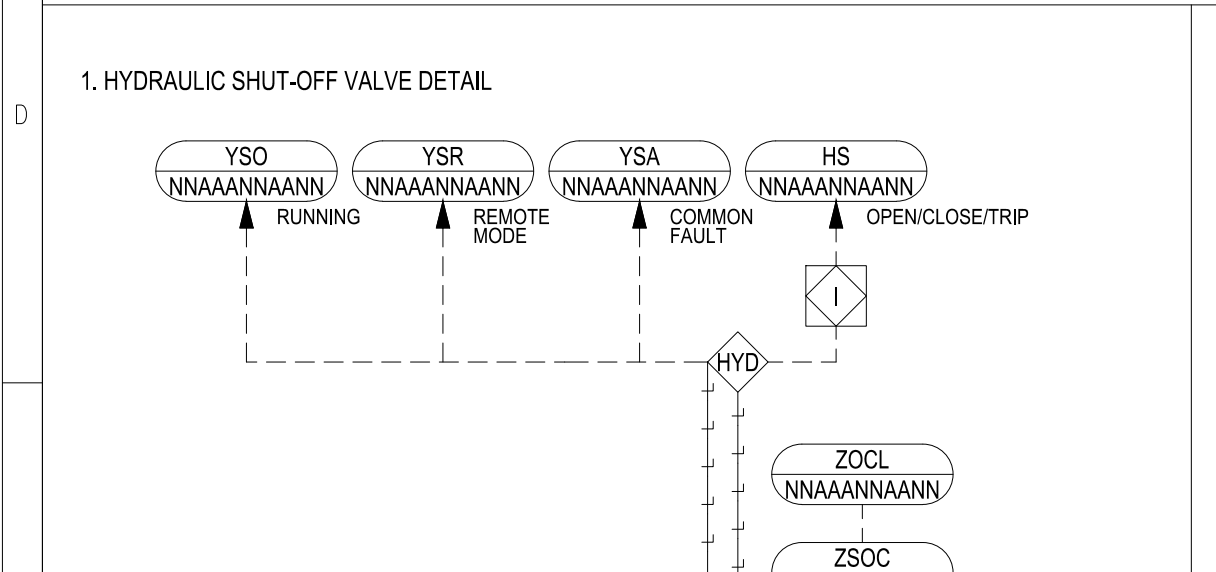
CONTROL VALVE TYPICAL FOR PNEUMATIC OPERATED VALVES	
4. PNEUMATIC CONTROL VALVE DETAIL (PNEUMATIC NON-RETURN VALVE DETAIL)	4. PNEUMATIC CONTROL VALVE TYPICAL



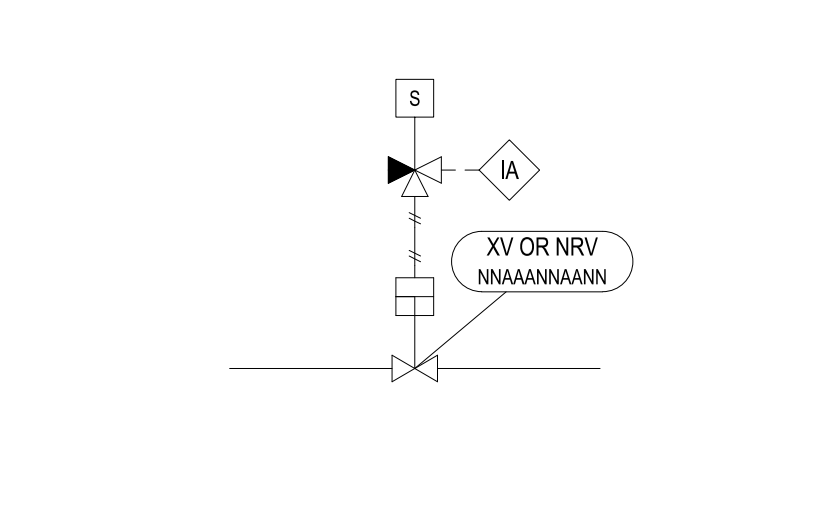
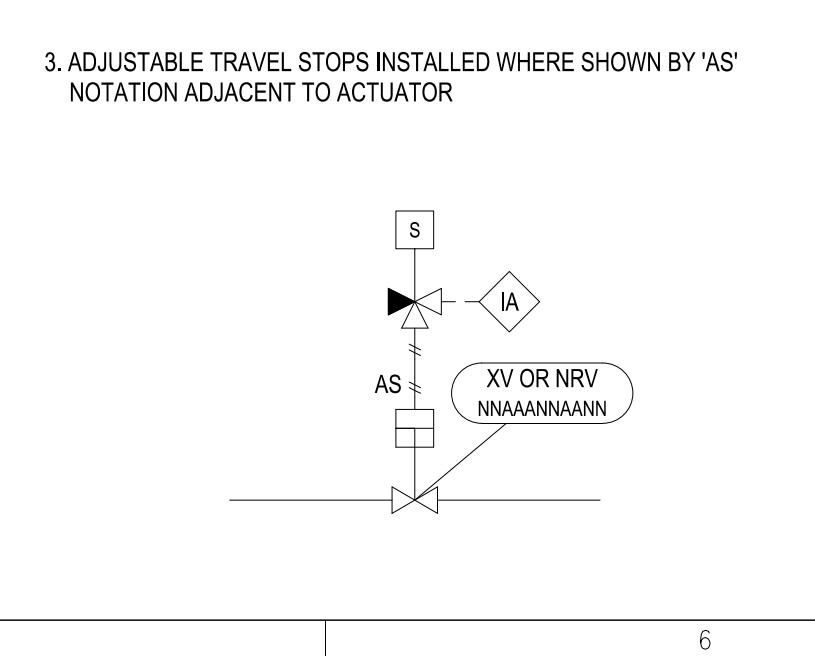
ABBREVIATION

ESDV : EMERGENCY SHUT-DOWN VALVE
SDV : SHUT-DOWN VALVE
SET : SET VALUE
ZSOC : POSITION SWITCH OPEN/CLOSE
ZIOC : POSITION INDICATOR OPEN/CLOSE
ZOCL : POSITION OPEN/CLOSE LAMP
CC : CATIONIC CONDUCTIVITY
SC : SPECIFIC CONDUCTIVITY
XV : SHUT-OFF VALVE
M : ELECTRIC MOTOR
MH : MAN HOLE
SMFCP : SETTLEMENT METERING FLOW COMPUTER PANEL
BMFCP : BACK-UP METERING FLOW COMPUTER PANEL
MECP : MAIN FIRE ALARM CONTROL PANEL
ESEW : EMERGENCY SHOWER & EYE WASHER
DCMS : DISTRIBUTED CONTROL AND MONITORING SYSTEM
BMS : BURNER MANAGEMENT SYSTEM
BPS : BOILER PROTECTION SYSTEM

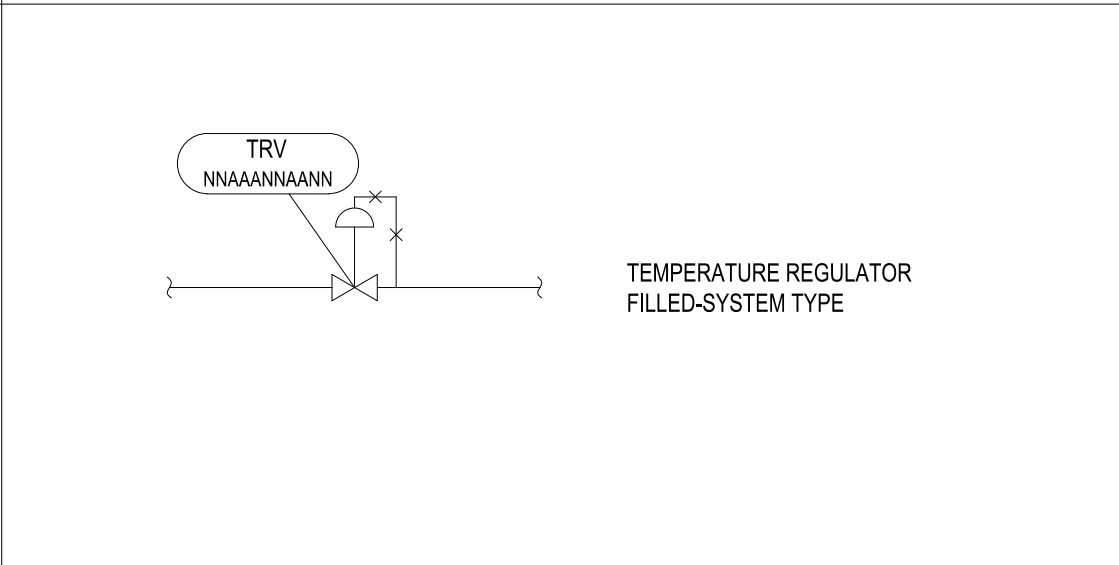
ON-OFF VALVE TYPICAL FOR HYDRAULIC AND PNEUMATIC OPERATED VALVES



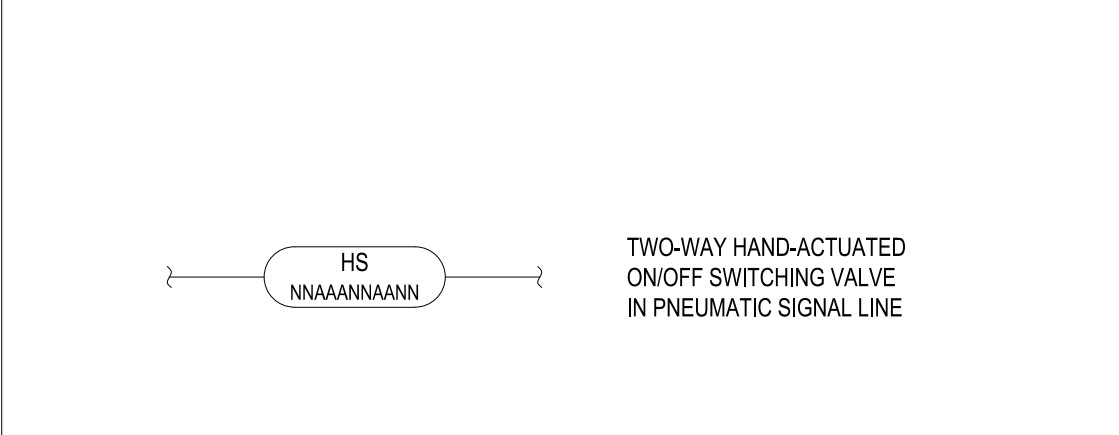
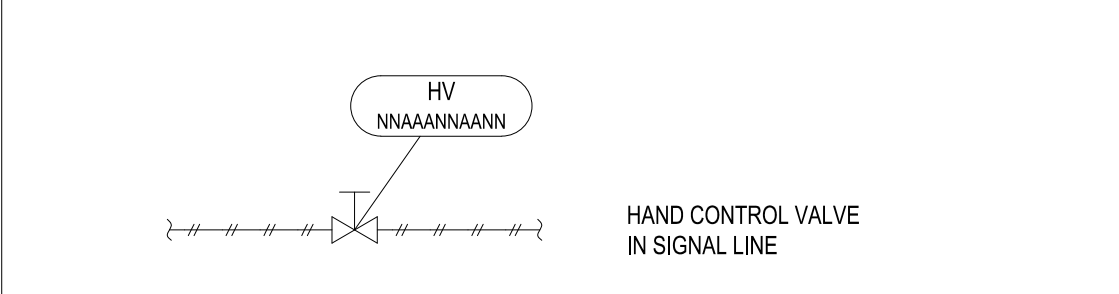
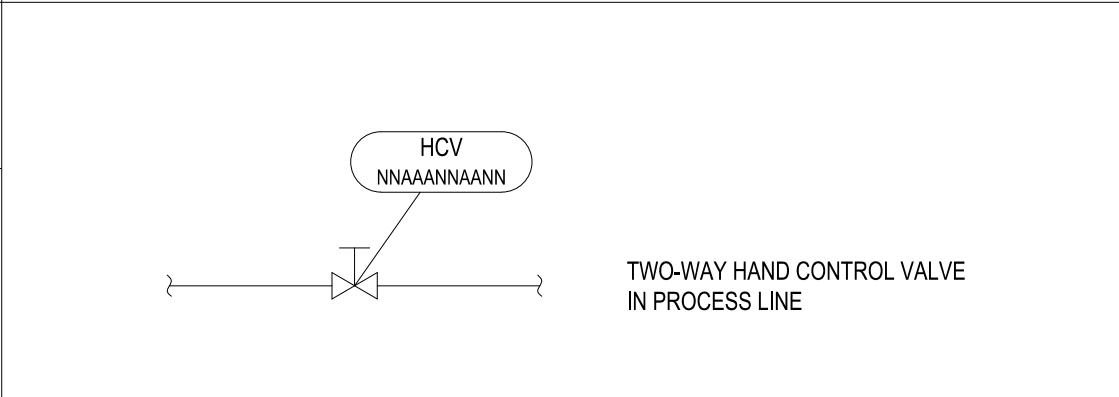
C		
	2. PNEUMATIC SHUT-OFF VALVE TYPICAL/PNEUMATIC NON-RETURN VALVE TYPICAL	

[illegible]

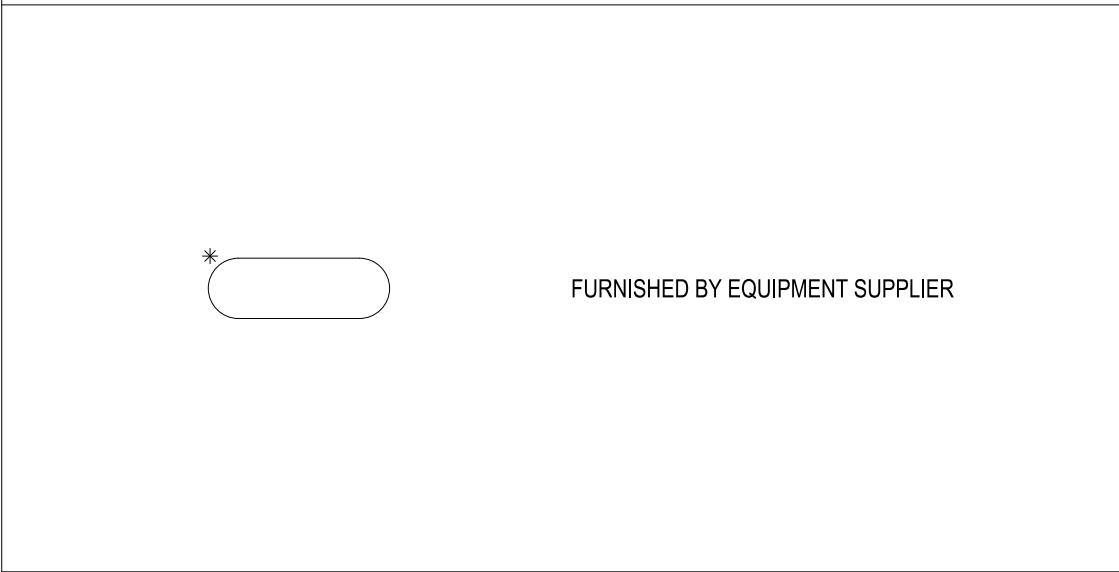
3	2
SELF-ACTUATED DEVICES-TEMP	



HAND ACTUATED DEVICES



SUPPLIER DESIGNATOR



LOGIC & SEQUENCE CONTROL

GENERAL

- INTERCONNECTING LOGIC OR SEQUENCE CONTROL

NORMALLY USED FOR LOCAL AUXILIARY CONTROL SYSTEM

NOT NORMALLY ACCESSIBLE TO OPERATOR

- PACKAGED PROGRAMMABLE LOGIC CONTROLLER OR DIGITAL LOGIC CONTROLS INTEGRAL TO THE DISTRIBUTED CONTROL EQUIPMENT

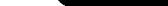
- NORMALLY ACCESSIBLE TO OPERATOR
- PACKAGED PROGRAMMABLE LOGIC CONTROLLER OR DIGITAL LOGIC CONTROLLER INTEGRAL TO THE DISTRIBUTED CONTROL EQUIPMENT

GENERAL NOTES :

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EMPLOYER					

TENDERER

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Engineering & Construction Group

PROJECT NAME	AL NOUF1 PROJECT
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DRAWING TITLE

P&ID FOR SYMBOLS & LEGENDS 3

A

PROJECT DWG NO.		SHEET	REV.
D00P-00GEN00-M05-0004		1/1	A
PROJECT NO.	REF. DRAWING NO.	SCALE	REF. CODE

D00P		NONE	
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EXAMPLE OF APPLICATION :

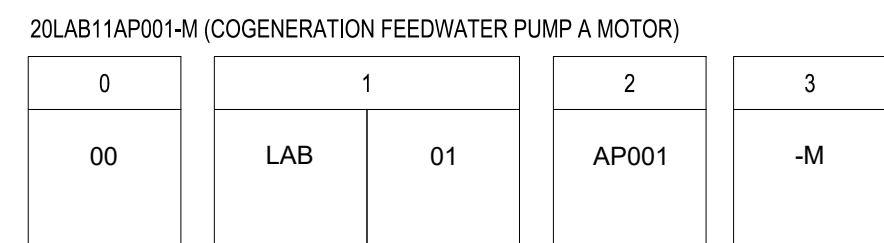
A	A	2
A	A	N
(1)	(2)	(3)

(3) SERIAL NO.

TAG NUMBERING SYSTEM IS BASED ON THE KKS NUMBERING SYSTEM. (DOC. NO.: 19K9-00GEN00-G02-0002)



EXAMPLE OF APPLICATION:

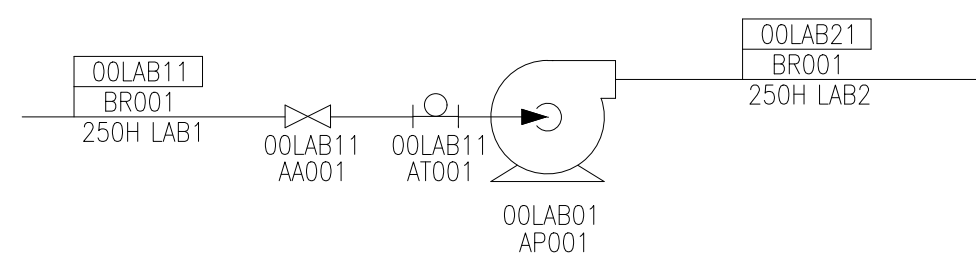


The diagram illustrates the structure of a Piping Design Code (PDC). It starts with a box labeled "PDC" containing "11LAB11" and "BR001". An arrow points from this box to a box labeled "BREAKDOWN LEVEL 0 & 1" containing "BREAKDOWN LEVEL 2". From this box, an arrow points to a box labeled "250H LAB1". From this box, an arrow points to the text "PROCESS DESIGN CODE". From this text, an arrow points to the text "PIPING INSULATION". From this text, an arrow points to a list of insulation types: "- H : INSULATION FOR HEAT CONSERVATION", "- P : INSULATION FOR PERSONAL PROTECTION", and "- N : NO INSULATION". From this list, an arrow points to the text "PIPING DIAMETER".

```

graph TD
    PDC["PDC  
11LAB11  
BR001"] --> BL["BREAKDOWN LEVEL 0 & 1  
BREAKDOWN LEVEL 2"]
    BL --> PDCODE["250H LAB1"]
    PDCODE --> PDCODE_LABEL["PROCESS DESIGN CODE"]
    PDCODE_LABEL --> PIPING_INSULATION["PIPING INSULATION"]
    PIPING_INSULATION --> INSULATION_TYPES["- H : INSULATION FOR HEAT CONSERVATION  
- P : INSULATION FOR PERSONAL PROTECTION  
- N : NO INSULATION"]
    INSULATION_TYPES --> PIPING_DIAMETER["PIPING DIAMETER"]
  
```

EXAMPLE FOR LINE VALVE, EQUIPMENT AND STRAINER IDENTIFICATION



NOTES :

100

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SAMSUNG

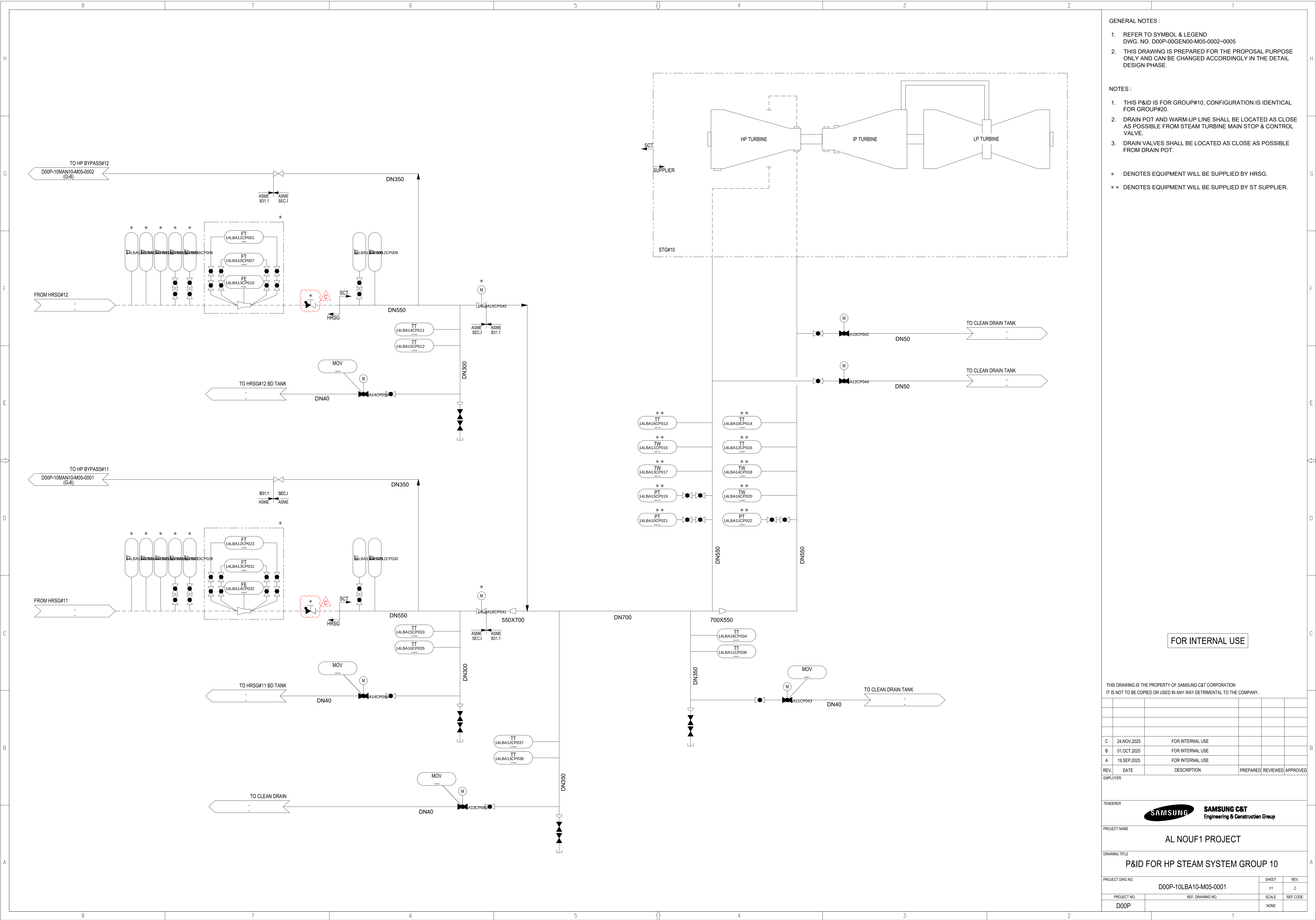
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PROJECT NAME	AL NOUF1 PROJECT
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DRAWING TITLE

P&ID FOR SYMBOLS & LEGENDS 4

100

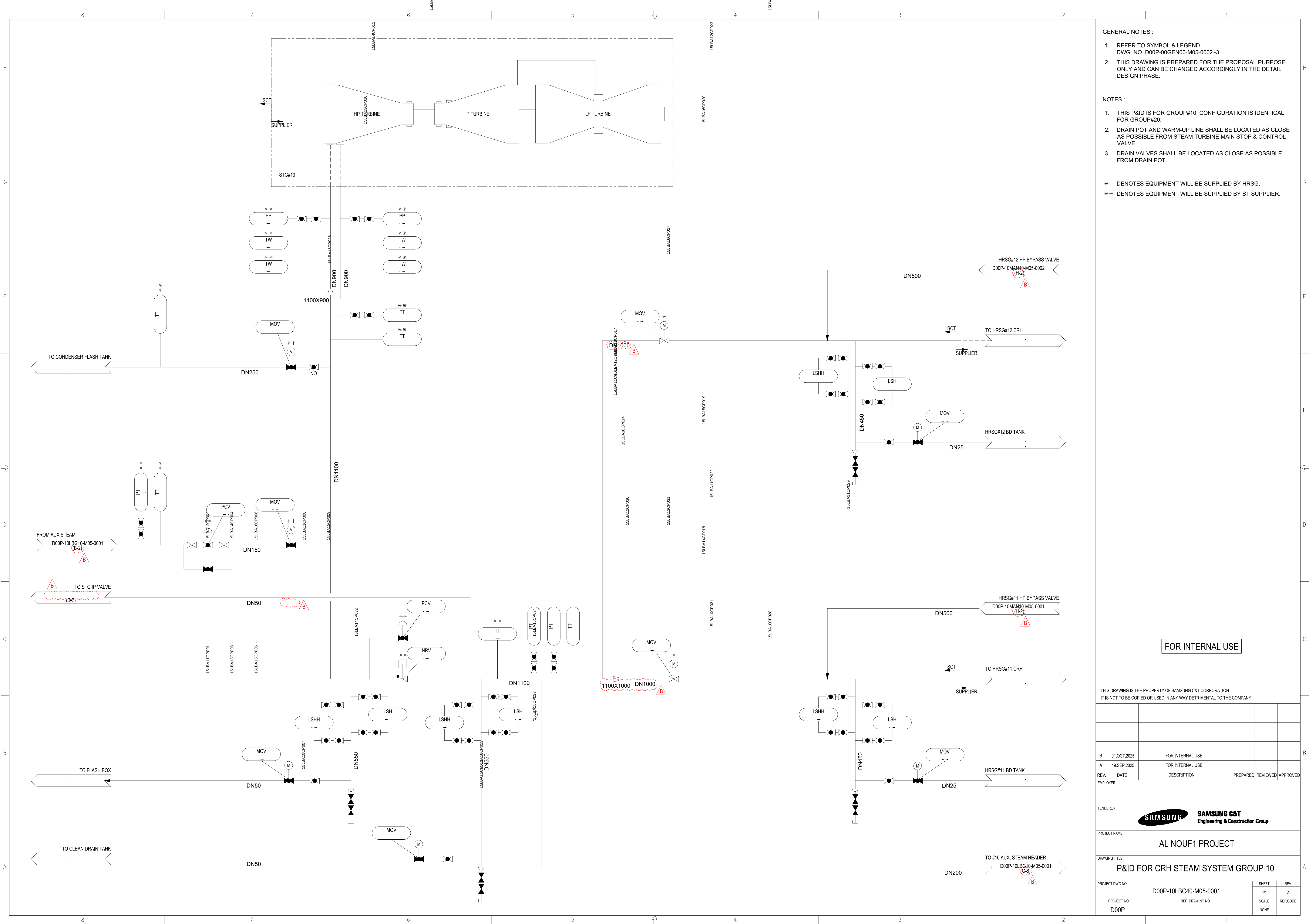


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DWG. NO. D00P-00GEN00-M05-0005
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- NOTES :
- THIS P&ID IS FOR GROUP#10, CONFIGURATION IS IDENTICAL FOR GROUP#20.
 - DRAIN POT AND WARM-UP LINE SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM STEAM TURBINE MAIN STOP & CONTROL VALVE.
 - DRAIN VALVES SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM DRAIN POT.
- * DENOTES EQUIPMENT WILL BE SUPPLIED BY HRSG.
** DENOTES EQUIPMENT WILL BE SUPPLIED BY ST SUPPLIER.

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REV.	DATE	DESCRIPTION	PREPARED	REVIEWED	APPROVED
EMPLOYER					
TENDERER					
PROJECT NAME					
AL NOUF1 PROJECT					
DRAWING TITLE					
P&ID FOR HP STEAM SYSTEM GROUP 10					
PROJECT DWG. NO.				SHEET	REV.
D00P-10LBA10-M05-0001				1/1	C
PROJECT NO.				SCALE	REF. CODE
D00P				NONE	



- GENERAL NOTES :
- REFER TO SYMBOL & LEGEND
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- NOTES :
- THIS P&ID IS FOR GROUP#10, CONFIGURATION IS IDENTICAL FOR GROUP#20.
 - DRAIN POT AND WARM-UP LINE SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM STEAM TURBINE MAIN STOP & CONTROL VALVE.
 - DRAIN VALVES SHALL BE LOCATED AS CLOSE AS POSSIBLE FROM DRAIN POT.
- * DENOTES EQUIPMENT WILL BE SUPPLIED BY HRSG.
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A	19.SEP.2025	FOR INTERNAL USE			
EMPLOYER					

TENDERER
SAMSUNG **SAMSUNG C&T**
Engineering & Construction Group

PROJECT NAME
AL NOUF1 PROJECT

DRAWING TITLE
P&ID FOR CRH STEAM SYSTEM GROUP 10

PROJECT DWG. NO.	PROJECT NO.	REF. DRAWING NO.	SHEET	REV.
D00P-10LBC40-M05-0001	D00P		1/1	A
			NONE	